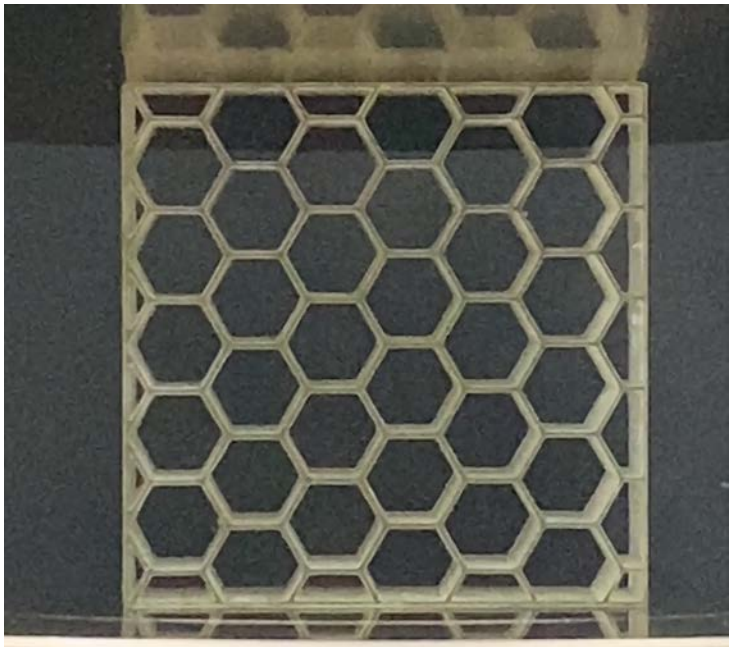
Mechanical Engineering Experiment Class

- Mechanics of Materials

연세대학교 기계공학부 김석

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1. Introduction
2. Concepts and theories regarding Tension Test
3. Experiment Procedure
4. Safety Guidelines
5. Requirements for pre/post Lab Report

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Introduction

Objectives

- Showing how stress can be related to strain by using experimental methods to determine the stress–strain diagram for a specific material.
- The behavior described by this diagram will then be discussed for materials that are commonly used in engineering.
- Mechanical properties and other tests that are related to the development of mechanics of materials will be discussed.

Experiment #1 – Tensile Test

[Learning objective]

➔ **Understand mechanical behavior of materials through the tensile test**

➔ **Characterize the mechanical properties of a known material**

➔ **Identify materials from the tensile tests**







Universal testing machine, AG-50kN XD Plus, Shimadzu co.

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Tension and Compression Test

- The strength of a material depends on its ability to sustain a load without undue deformation or failure
- The mechanical property is inherent in the material itself and must be determined by experiment
- One of the most important tests is the tension or compression test
- From the test, the relationship between the average normal stress and average normal strain was obtained

Mechanical Properties

Appendix B. Typical Properties of Selected Materials Used in Engineering[1,5] (SI Units)

Material	Density kg/m³	Ultimate Strength: Tension, MPa	Ultimate Strength: Compression,[2] MPa	Ultimate Strength: Shear, MPa	Yield Strength[3]: Tension, MPa	Yield Strength[3]: Shear, MPa	Modulus of Elasticity, GPa	Modulus of Rigidity, GPa	Coefficient of Thermal Expansion, 10⁻⁶/°C	Ductility, Percent Elongation in 50 mm
Steel										
Structural (ASTM-A36)	7860	400			250	145	200	77.2	11.7	21
High-strength-low-alloy										
ASTM-A709 Grade 345	7860	450			345		200	77.2	11.7	21
ASTM-A913 Grade 450	7860	550			450		200	77.2	11.7	17
ASTM-A992 Grade 345	7860	450			345		200	77.2	11.7	21
Quenched & tempered										
ASTM-A709 Grade 690	7860	760			690		200	77.2	11.7	18
Stainless, AISI 302										
Cold-rolled	7920	860			520		190	75	17.3	12
Annealed	7920	655			260	150	190	75	17.3	50
Reinforcing Steel										
Medium strength	7860	480			275		200	77	11.7	
High strength	7860	620			415		200	77	11.7	
Cast Iron										
Gray Cast Iron										
4.5% C, ASTM A-48	7200	170	655	240			69	28	12.1	0.5
Malleable Cast Iron										
2% C, 1% Si, ASTM A-47	7300	345	620	330	230		165	65	12.1	10
Aluminum										
Alloy 1100-H14										
(99% Al)	2710	110		70	95	55	70	26	23.6	9
Alloy 2014-T6	2800	455		275	400	230	75	27	23.0	13
Alloy-2024-T4	2800	470		280	325		73		23.2	19
Alloy-5456-H116	2630	315		185	230	130	72		23.9	16
Alloy 6061-T6	2710	260		165	240	140	70	26	23.6	17
Alloy 7075-T6	2800	570		330	500		72	28	23.6	11
Copper										
Oxygen-free copper (99.9% Cu)										
Annealed	8910	220		150	70		120	44	16.9	45
Hard-drawn	8910	390		200	265		120	44	16.9	4
Yellow-Brass (65% Cu, 35% Zn)										
Cold-rolled	8470	510		300	410	250	105	39	20.9	8
Annealed	8470	320		220	100	60	105	39	20.9	65
Red Brass (85% Cu, 15% Zn)										
Cold-rolled	8740	585		320	435		120	44	18.7	3
Annealed	8740	270		210	70		120	44	18.7	48
Tin bronze (88 Cu, 8Sn, 4Zn)	8800	310			145		95		18.0	30
Manganese bronze (63 Cu, 25 Zn, 6 Al, 3 Mn, 3 Fe)	8360	655			330		105		21.6	20
Aluminum bronze (81 Cu, 4 Ni, 4 Fe, 11 Al)	8330	620	900		275		110	42	16.2	6

(Table continued on page 747)

Appendix B. Typical Properties of Selected Materials Used in Engineering[1,5] (SI Units)

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Magnesium Alloys										
Alloy AZ80 (Forging)	1800	345		160	250		45	16	25.2	6
Alloy AZ31 (Extrusion)	1770	255		130	200		45	16	25.2	12
Titanium										
Alloy (6% Al, 4% V)	4730	900			830		115		9.5	10
Monel Alloy 400(Ni-Cu)										
Cold-worked	8830	675			585	345	180		13.9	22
Annealed	8830	550			220	125	180		13.9	46
Cupronickel (90% Cu, 10% Ni)										
Annealed	8940	365			110		140	52	17.1	35
Cold-worked	8940	585			545		140	52	17.1	3
Timber, air dry										
Douglas fir	470	100	50	7.6			13	0.7	Varies	
Spruce, Sitka	415	60	39	7.6			10	0.5	3.0 to 4.5	
Shortleaf pine	500		50	9.7			12			
Western white pine	390		34	7.0			10			
Ponderosa pine	415	55	36	7.6			9			
White oak	690		51	13.8			12			
Red oak	660		47	12.4			12			
Western hemlock	440	90	50	10.0			11			
Shagbark hickory	720		63	16.5			15			
Redwood	415	65	42	6.2			9			
Concrete										
Medium strength	2320		28				25		9.9	
High strength	2320		40				30		9.9	
Plastics										
Nylon, type 6/6, (molding compound)	1140	75	95		45		2.8		144	50
Polycarbonate	1200	65	85		35		2.4		122	110
Polyester, PBT (thermoplastic)	1340	55	75		55		2.4		135	150
Polyester elastomer	1200	45		40			0.2			500
Polystyrene	1030	55	90		55		3.1		125	2
Vinyl, rigid PVC	1440	40	70		45		3.1		135	40
Rubber	910	15							162	600
Granite (Avg. values)	2770	20	240	35			70	4	7.2	
Marble (Avg. values)	2770	15	125	28			55	3	10.8	
Sandstone (Avg. values)	2300	7	85	14			40	2	9.0	
Glass, 98% silica	2190		50				65	4.1	80	

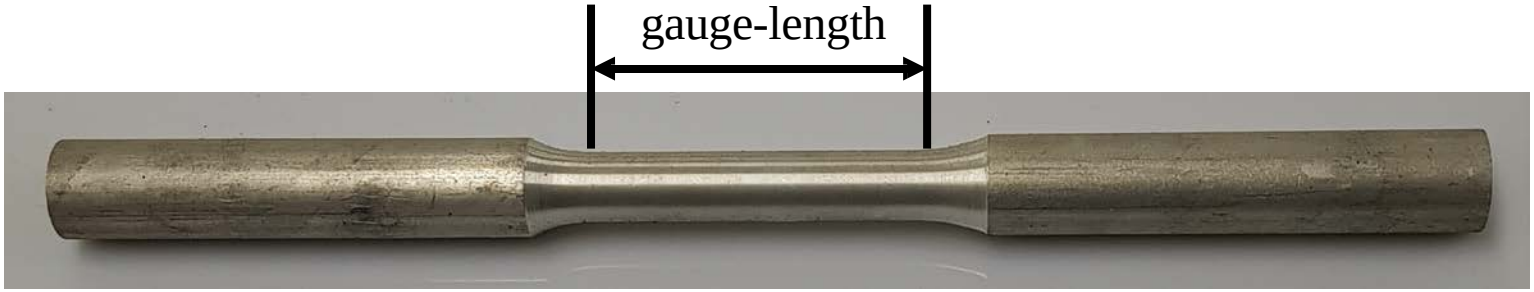
[1]Proprties of metals very widely as a result of variations in composition, heat treatment, and mechanical working.
[2]For ductile metals the compression strength is generally assumed to be equal to the tension strength.
[3]Offset of 0.2 percent.
[4]Timber proprties are for loading parallel to the grain.
[5]See also *Marks' Mechanical Engineering Handbook*, 10th ed., McGraw-Hill, New York, 1996; *Annual Book of ASTM*, American Society for Testing Materials, Philadelphia, Pa.; *Metals Handbook*, American Society of Metals, Metals Park, Ohio; and *Aluminum Design Manual*, The Aluminum Association, Washington, DC.

Mechanics of materials by Ferdinand P. Beer, 5th Edition

Test Specimen

A specimen of the material for test is made into a "standard" shape and size

- **A specimen has a constant circular cross section with enlarged end, so that failure will not occur at the grips**
- **Before testing, two small punch marks are placed along the specimen's uniform length.**
- **Measurements are taken of both the specimen's initial cross-sectional area, and the *gauge-length* distance between the punch marks.**



Aluminum specimen

Tension and Compression Test Machine

A testing machine is then used to stretch the specimen at a very slow, constant rate until it fails.

- During the test, data is recorded of the applied load *P*
- Measuring the elongation between the marks on the specimen using either a caliper or a device called an extensometer. The elongation is then used to calculate the average normal strain in the specimen.
- Reading the strain directly by using an electrical-resistance strain gauge in the figure below. By measuring the electrical resistance of the wire, the gauge may be calibrated to read values of normal strain directly.



Testing machine

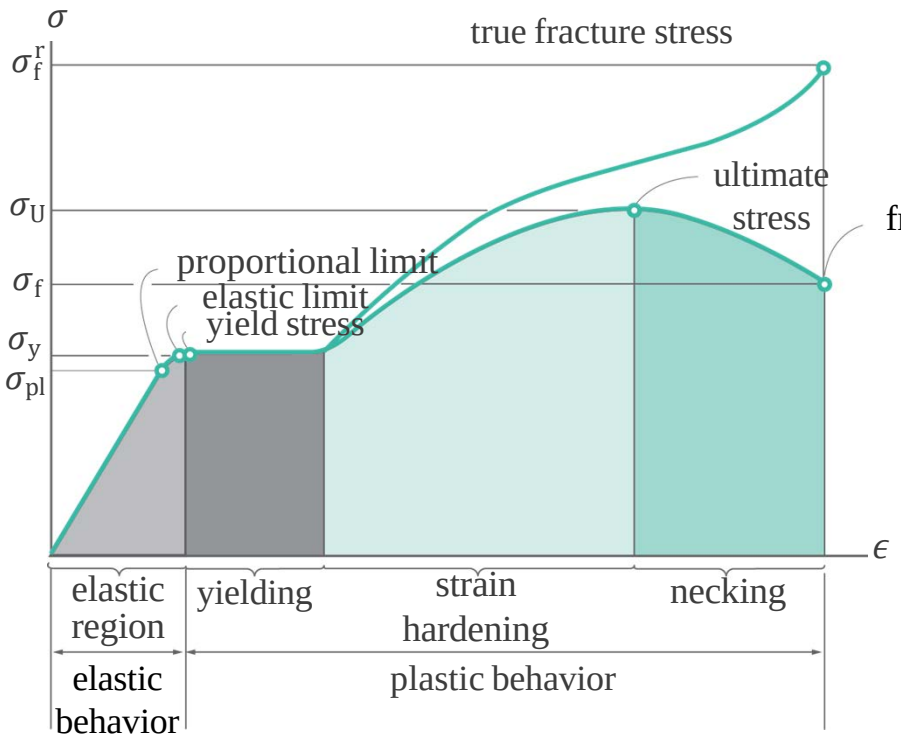
Electrical–resistance strain gauge



Extensometer

The Stress-Strain Diagram

Conventional and true stress-strain diagrams for ductile material (steel)



✓ Nominal Stress vs ✓ True Stress

(=engineering stress)

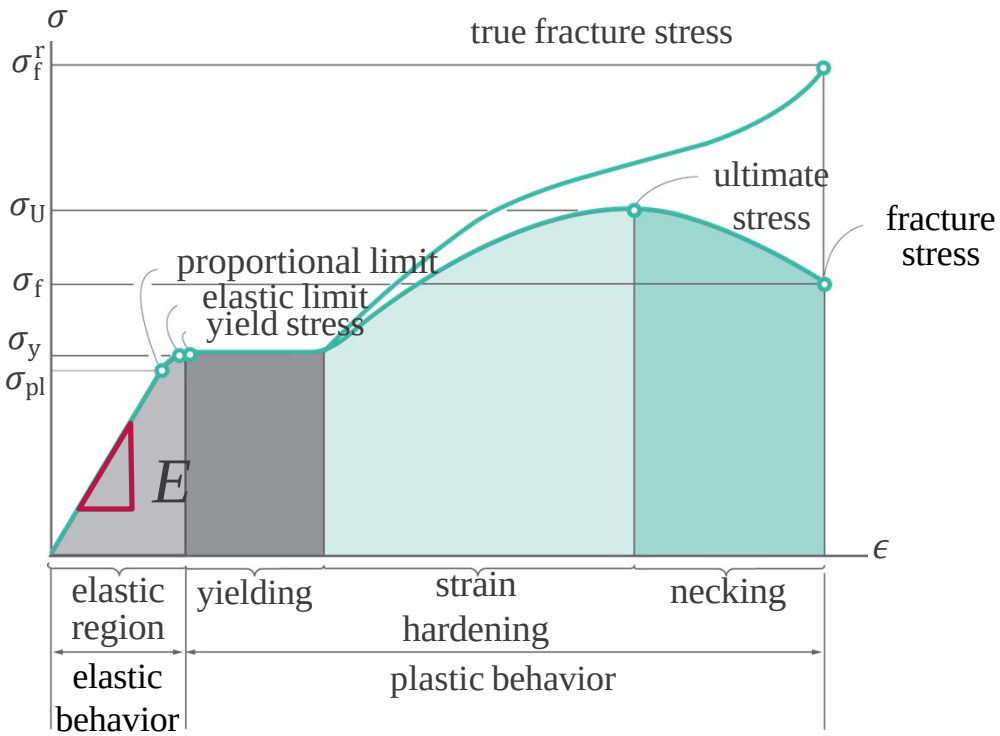
$$\sigma_{\text{eng}} = \frac{P}{A_0} \qquad \sigma_{\text{true}} = \frac{P}{A}$$

✓ Nominal Strain (=engineering strain) vs ✓ True Strain

$$\varepsilon_{\text{eng}} = \frac{\delta}{L_0} \qquad \varepsilon_{\text{true}} = \sum \frac{\delta L_k}{L_k}$$

$$= \int_{L_0}^{L} \frac{dL}{L} = \ln \frac{L}{L_0}$$

Linear elastic region - **Hooke's Law**



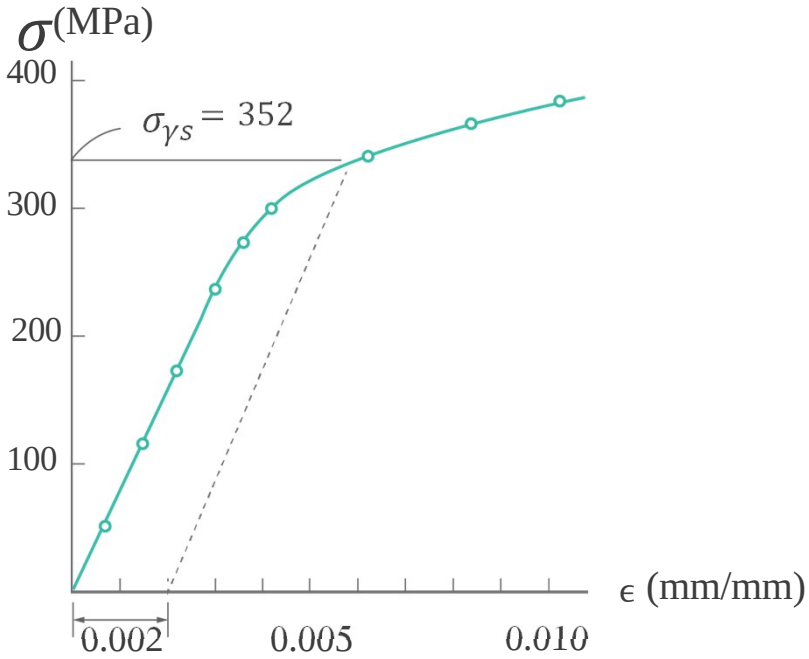
In elastic regime, an increase in stress causes a proportionate increase in strain. *Linear relationship* between stress and strain can is described by *Hooke's law*.

$$\sigma = E\varepsilon$$

E : modulus of elasticity or Young's modulus

⇒ A measure of how stiff a material is

Offset Method



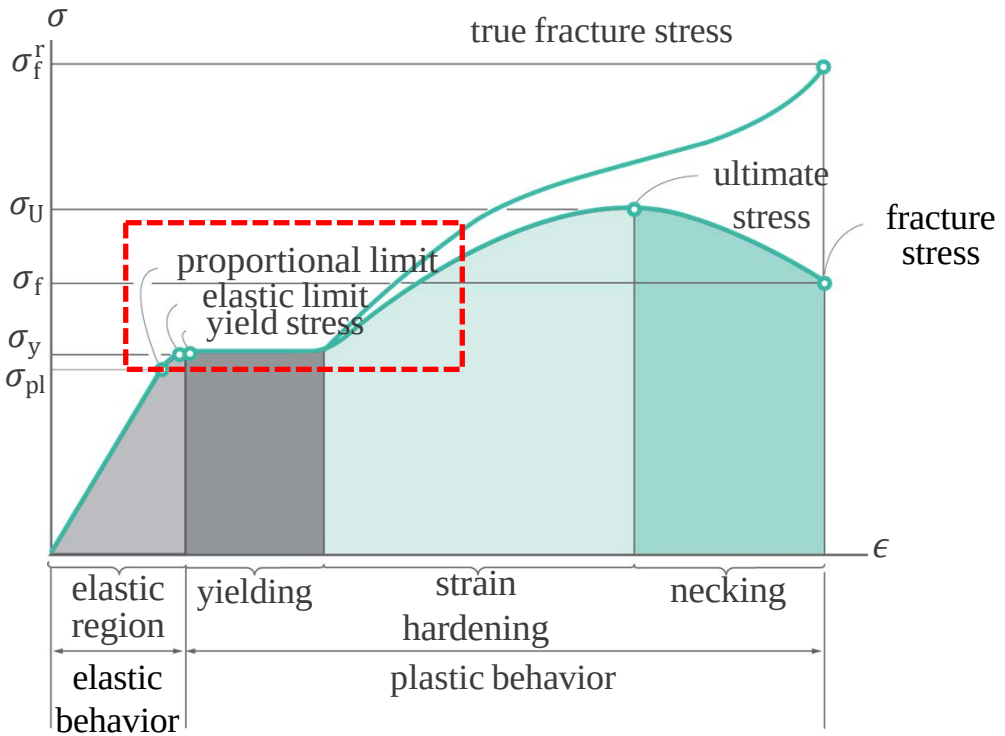
(0.2% offset)

Yield strength for an aluminum alloy

Offset method

Some ductile materials, e.g. Al, do not show a clear yielding point.

Non-linear region

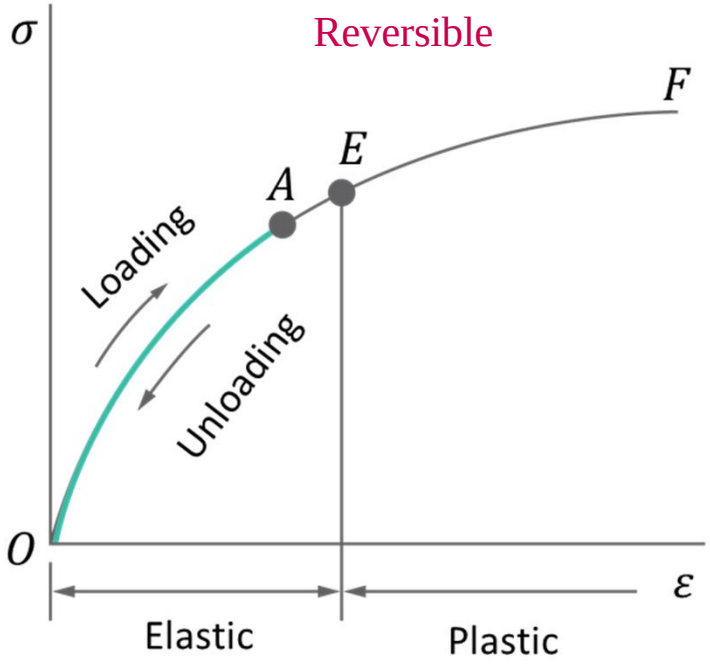


Material Strength

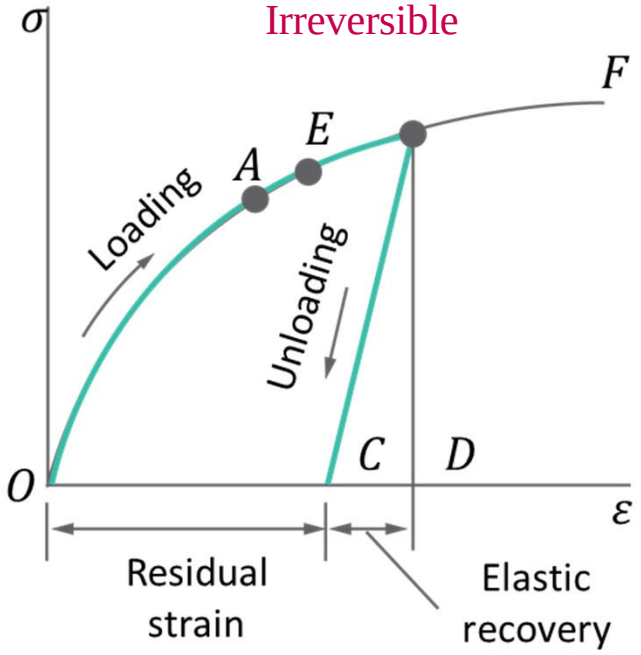
- proportional limit (σ_{pl})
- elastic limit
- yield strength (σ_Y)
- ultimate strength (σ_u)
- fracture strength (σ_f)

Elastic vs non-Elastic(or plastic)

Reversible



Irreversible

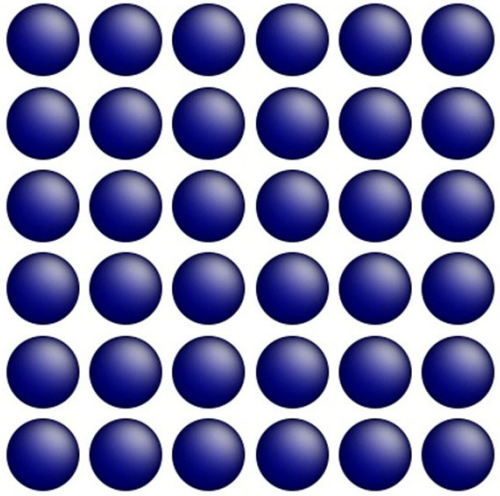


During unloading, if the specimen returns to its original dimensions, it is called elastic. If not, it undergoes plasticity (or permanent deformation)

During unloading after yielding, the loading curve follows the initial tangent line of stress-strain curve, leading to residual strain after entirely removing the load

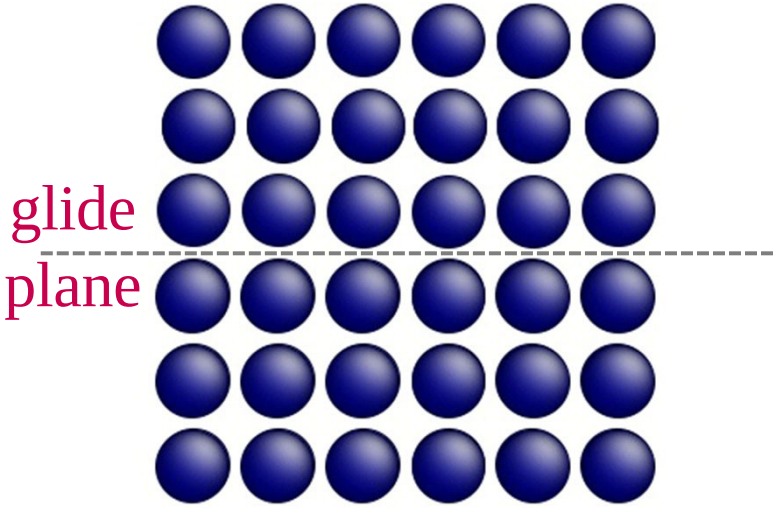
Elastic vs non-Elastic(or plastic)

Elastic Deformation



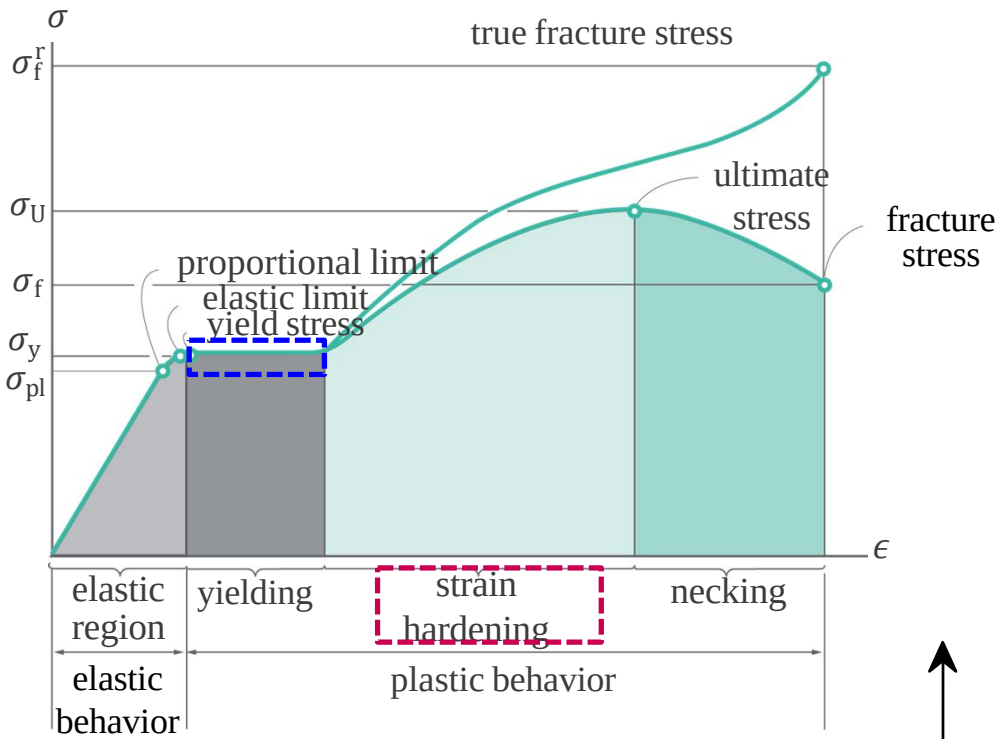
No bond breaking

Plastic Deformation



Bond breaking, rebonding

Post-yielding Behavior

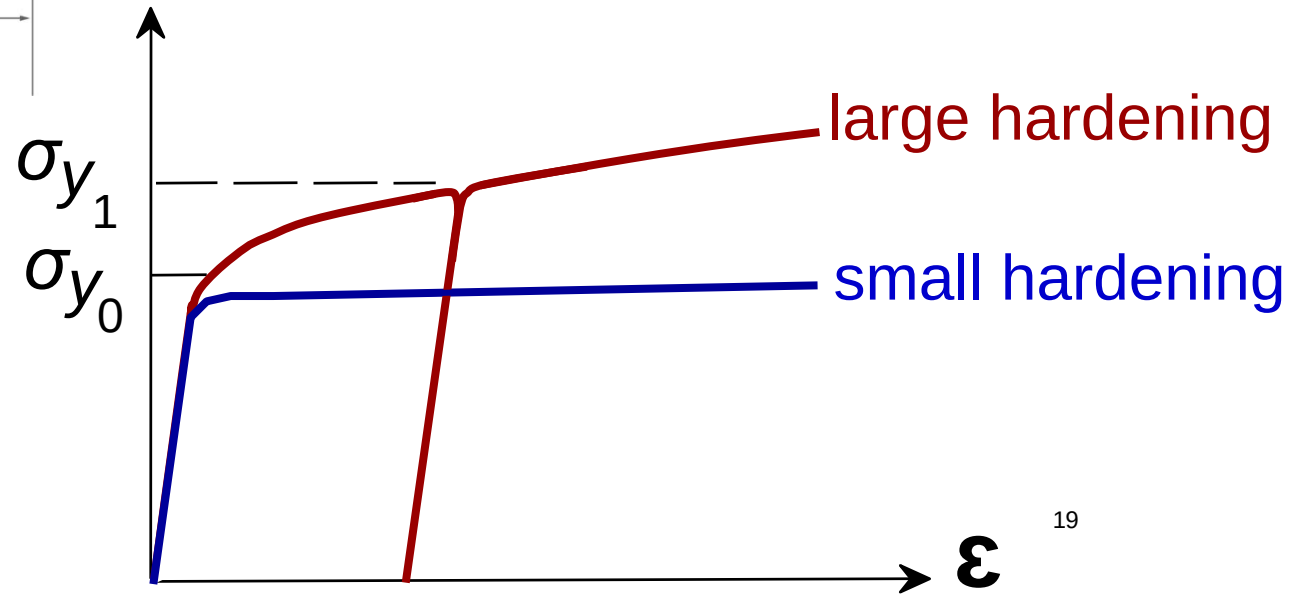


Perfectly Plastic

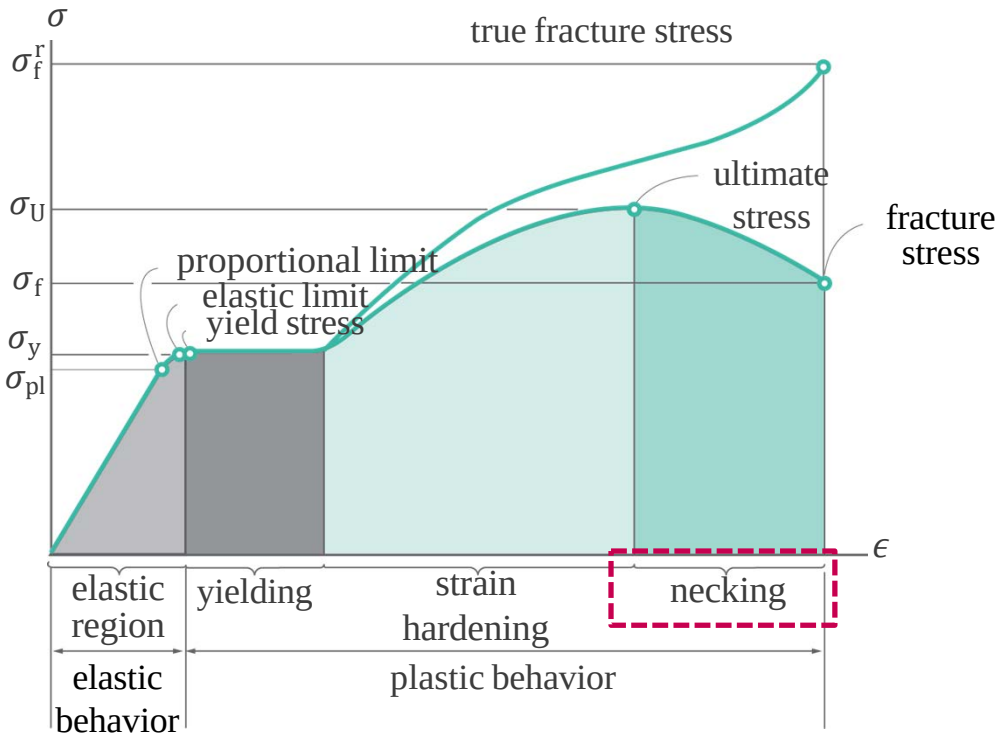
After yielding, the material deforms without an increases in the applied force

Strain Hardening (or Work Hardening)

Material shows hardening behavior, which requires increased stress for further deformation.

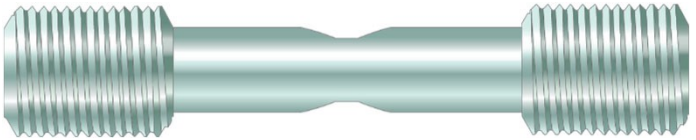


Necking



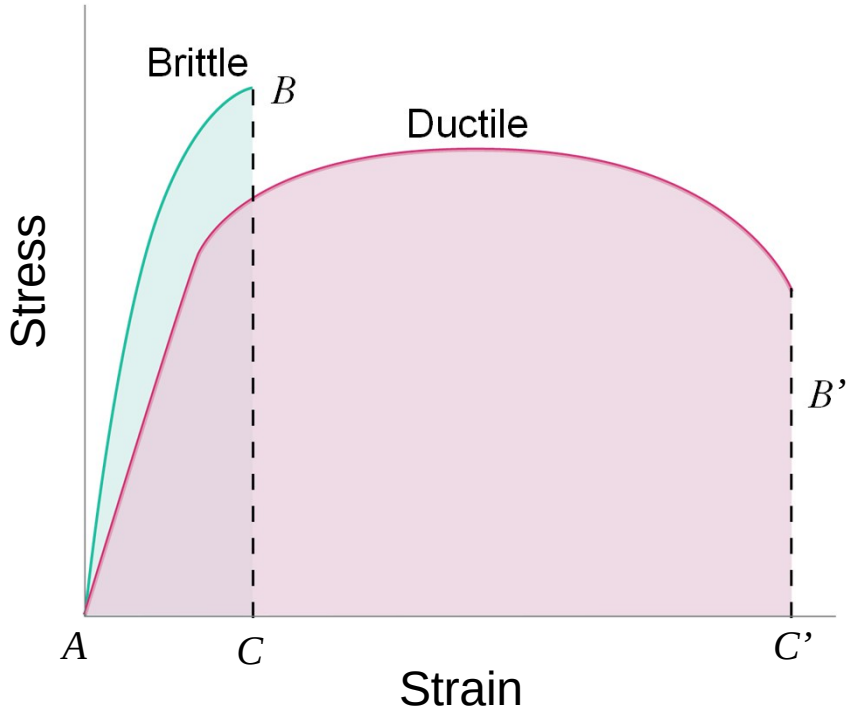
The reduction in area of the bar becomes clearly visible.

The reduction of (engineering) stress is due to the decrease in area of the bar and not to a loss in strength of the material itself.



Necking

Fracture



Schematic representation of tensile behavior of brittle and ductile materials

Materials that can be subjected to large strain before fracture are capable of absorbing large amounts of strain energy prior to fracture.

e.g. aluminum, copper, magnesium, lead, molybdenum, nickel, brass, bronze, monel metal (Ni-Cu alloy), nylon, teflon

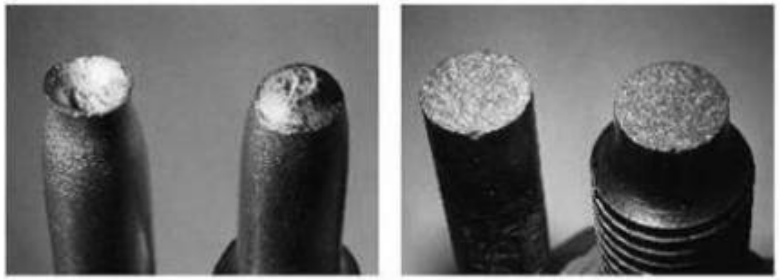
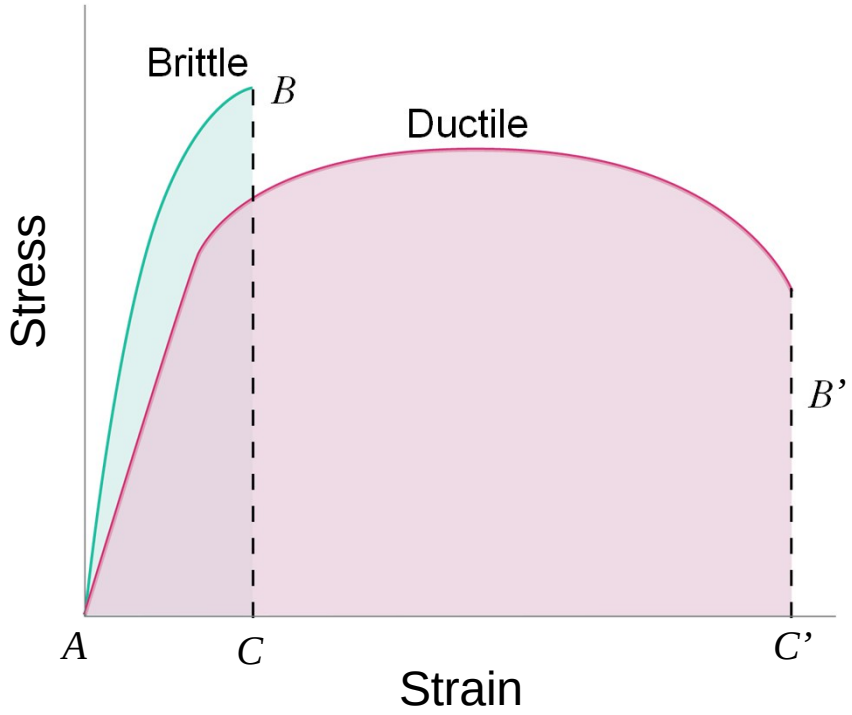


Image source: https://practicalmaintenance.net/?p=1135

The formation of a “cup-cone” shape at the fracture location, which is characteristic of ductile materials.

Fracture



Schematic representation of tensile behavior of brittle and ductile materials

Materials that can be subjected to large strain before fracture are capable of absorbing large amounts of strain energy prior to fracture.

Modulus of Toughness

- The modulus of toughness represents the entire area under the stress–strain diagram.
- It indicates the strain-energy density of the material just before it fractures.

$$U_r = \int_0^{\epsilon_y} \sigma \, d\epsilon \ (\mathrm{J/m^3})$$

- Toughness of a material is expressed in several contexts
- For this lecture, toughness = **amount of energy absorbed before fracture**
- Approximate by area under the stress-strain curve—units of energy per unit volume

$\mathrm{J/m^3}$

Percent Elongation and Percent Reduction in Area

- Ductile Materials
  - Any material that can be subjected to large strains before it fractures is called a **ductile material**.
  - These materials are capable of absorbing shock or energy, and if they become overloaded, they will usually exhibit large deformation before failing.
  - One way to specify the ductility of a material is to report its percent elongation or percent reduction in area at the time of fracture.

  - The ***percent elongation :*** $\frac{L_f - L_0}{L_0}$

  - The ***percent reduction in area :*** $\frac{A_0 - A_f}{A_0}$

Materials Properties Obtainable from Tension Tests

Category	Engineering Property	True Stress–Strain Property
Elastic constants	Elastic modulus, E, E_t Poisson's ratio, ν	—
Strength	Proportional limit, σ_p Yield strength, σ_o Ultimate tensile strength, σ_u Engineering fracture strength, σ_f	True fracture strength, $\tilde{\sigma}_{fB}$ Strength coefficient, H
Ductility	Percent elongation, $100\varepsilon_f$ Reduction in area, $\%RA$	True fracture strain, $\tilde{\varepsilon}_f$
Energy capacity	Tensile toughness, u_f	True toughness, $\tilde{u}_f$
Strain hardening	Strain hardening ratio, σ_u/σ_o	Strain hardening exponent, n

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Experiment Procedure

Check size of specimen



Check
Fixture Alignment



Gripping (Crosshead)
Specimen



Extensometer Installation



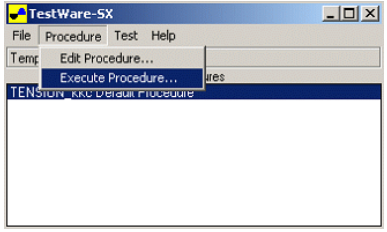
Gripping(Actuator)
Specimen



Force & Strain Zero Control



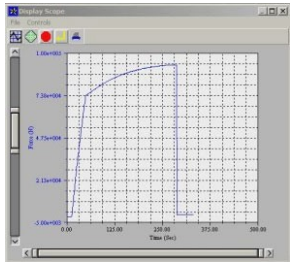
Test Run(Control PC)



Check Data and fracture



Save Test Data



Remove Extensometer

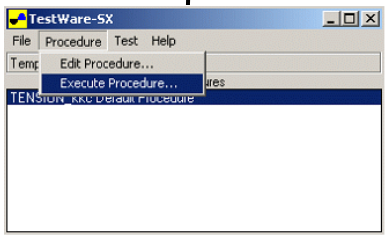


Un-Gripping Specimen



Check
Young's Modulus E

Test Stop(Control PC)



Data Convert(σ-ε Curve)

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Safety Guidelines

- **Keep hands, head, and all other body parts away from the testing space when the crosshead is in motion**
- **Moving parts can produce dangerous forces corresponding to the system capacity. All operators should read the instruction manual thoroughly to ensure they operate the system correctly. Control access to the system to ensure only trained operators operate the system**
- **Due to risk of electric shock, never touch inside the controller unit, located in the testing machine**
- **Do not open the ball screw shaft protective cover unless lubricating the screw shaft. Do not leave the shaft cover open while performing tests. Keep hair, clothing, rags, and fingers from getting caught in the ball screw when lubricating the ball screw**
- **Keep the face and body away from the test specimen during tests. When the specimen breaks, flying debris could injure eyes or other body parts. Always use a safety guard or other measure to provide protection from broken specimens**
- **Due to risk of injury, or fire, wiring, maintenance, and inspection work should be performed only by a person with specialized expertise.**
- **Do not operate the system in a manner that subjects the crosshead to loads when the load cell is removed and do not operate the system in a manner that does not transmit the load to the load cell. Doing so would cause overloading to be undetected, which could result in damage to the frame or jigs**
- **Never operate the crosshead without crosshead limit switches in place. Doing so could cause the crosshead to hit the jig and damage the frame, jig, or load cell**
- **Before setting the input voltage, be sure to turn OFF the power switch and disconnect the power supply cable**

Safety Guidelines

- 교두대가 움직일 때, 손, 머리 및 모든 신체 부위를 시험 공간에서 멀리 두십시오.
- 움직이는 부품은 시스템 용량에 해당하는 위험한 힘을 생성할 수 있습니다.

  모든 운영자는 시스템을 올바르게 작동하기 위해 설명서를 철저히 읽어야 합니다. 시스템에 대한 접근을 통제하여 훈련된 운영자만이 작동할 수 있도록 하십시오.
- 감전 위험이 있으므로 시험 장비에 위치한 컨트롤러 유닛을 절대 만지지 마십시오.
- 볼스크류 샤프트에 윤활유를 바르지 않는 한, 샤프트 보호 덮개를 열지 마십시오. 시험을 수행하는 동안 샤프트 덮개를 열어 두지 마십시오.

  볼스크류에 윤활유를 바를 때, 머리카락, 옷, 천 및 손가락이 볼스크류에 끼이지 않도록 주의하십시오.

안전규칙

- 시험 중에는 얼굴과 신체를 시험 시편에서 멀리 두십시오. 시편이 파손될 때, 튀는 파편이 눈이나 다른 신체 부위를 다치게 할 수 있습니다. 항상 안전 가드 또는 기타 보호 조치를 사용하여 파손된 시편으로부터 보호하십시오.
- 부상이나 화재의 위험이 있으므로 배선, 유지보수 및 점검 작업은 반드시 전문 지식을 갖춘 사람만이 수행해야 합니다.
- 로드셀을 제거한 상태에서 교두대에 하중이 가해지도록 시스템을 작동하거나,하중이 로드셀로 전달되지 않는 방식으로 시스템을 작동하지 마십시오.

  이런 경우 과부하가 감지되지 않아 프레임이나 지그가 손상될 수 있습니다.
- 교두대 제한 스위치가 없는 상태에서 교두대를 작동하지 마십시오.

  그런 경우 교두대가 지그를 치고 프레임, 지그, 로드셀 등이 손상될 수 있습니다.
- 입력 전압을 설정하기 전에 반드시 전원 스위치를 끄고 전원 공급 케이블을 분리하십시오.

Contents

1. Introduction
2. Concepts and theories regarding Tension Test
3. Experiment Procedure
4. Safety Guidelines
5. Requirements for pre/post Lab Report

Pre-Lab Report

1. Explain why the gauge length and cross-sectional shape of a tensile test specimen are essential.
2. Using AI tools, develop an AI-assisted stress–strain curve simulator and compare the stress–strain behavior of the following five materials. Use the material properties provided. Using the simulator, perform tensile tests for all five materials and plot the stress–strain curves for all materials on a single graph.
3. Suppose a hypothetical material whose Young's modulus and yield strength are 50% and 30% greater, respectively, than those of Aluminum Alloy 2024-T4. Plot the s–s curves for both materials on the same graph. Discuss the significance of these changes from the perspective of material properties.
4. Using the simulator, plot the stress–strain curves for cold-worked and annealed Monel Alloy 400 (Ni-Cu) on a single graph. Explain why the curves exhibit different mechanical behavior, considering the effects of the processing history.
5. Discuss safety precautions and potential sources of error when performing a tensile test using a UTM.

Post-Lab Report

1. Plot a stress-strain curve using the measured load-displacement data.
2. Determine the Young’s modulus from the stress-strain curve using the measured data. And explain mechanical properties (such as yield stress, fracture stress, Poisson’s ratio, and so on).
3. Describe the yield, strain hardening, necking, and fracture behavior of the measured material based on its properties. And explain why a loud sound is generated at the moment the specimen fractures.
4. Compare the tensile and compression tests, and explain how their mechanical properties (e.g., Young's modulus, strength) differ.
5. Suggest ways to improve the experiment to reduce sources of error or obtain more information.